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Human Anatomy

A Comprehensive Illustrated Study Guide

Foundations, skeletal, muscular, circulatory & integumentary systems

An original study guide by the EverythingABUAD team.

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HOW TO USE THIS GUIDE

Each section opens with a short orientation paragraph, then builds up detail in prose and tables. **Bold** marks key terms worth memorising; shaded callout boxes flag high-yield facts, common traps and clinical correlates. Comparison tables are designed to be revised directly — cover one column and recall it from the other.

PART ONE · FOUNDATIONS OF ANATOMY

1. Introduction to Anatomy

Human anatomy is the branch of biological science devoted to the structure of the body and to the way that structure supports function. The word itself comes from the Greek for **"to cut up,"** a reminder that the discipline grew out of dissection: carefully separating the body into its parts to see how they are put together. A guiding principle runs through the whole subject — the form of every structure, from a single cell to an entire organ, is adapted to the job (or jobs) it must perform. Learning anatomy well therefore means learning to read structure as a clue to function.

1.1 Disciplines within anatomy

Because the body can be examined at very different scales and stages, anatomy is divided into several complementary disciplines. Each looks at the same body through a different lens.

Discipline	What it studies
Gross (macroscopic) anatomy	Structures large enough to see with the unaided eye — also called topographic anatomy.
Microscopic anatomy (histology)	Tissues and cells as seen under the microscope, below the limit of naked-eye vision.
Developmental anatomy (embryology)	How the body forms and changes from conception through growth.
Neuroanatomy	The structure of the brain, spinal cord and the rest of the nervous system.

1.2 Approaches to studying anatomy

The same body of knowledge can be organised in more than one way, and each arrangement suits a different purpose. Three approaches are used routinely, with a fourth that is often overlooked.

Regional (topographic) approach

Here the body is studied one region at a time — the upper and lower limbs, the thorax and abdomen, the pelvis and perineum, the head and neck, the back, and so on — examining every structure that lies within the region together. This approach works especially well when it is paired with hands-on dissection, but it can obscure how a single system runs continuously through the whole body.

Systemic approach

Here one body system is followed from head to toe before moving on to the next. If the cardiovascular system is chosen, for example, the heart and the entire network of blood vessels are studied as a unit. Systems studied this way include the skeletal system (osteology), the articular system (the joints), the reproductive system, and the rest. This guide is built mainly around the systemic approach.

Clinical approach

Here the emphasis falls on the anatomy that matters most in practice — in medicine, dentistry and the allied health sciences — selecting and stressing the structures a clinician must know. Each approach has its own strengths and weaknesses, so they are best used together rather than in isolation.

> **THE FOURTH, OFTEN-FORGOTTEN APPROACH: SURFACE ANATOMY**
> Surface anatomy studies the living body at rest and in motion, learning to picture the structures that lie just beneath the skin and can be felt through it. It underpins routine clinical skills such as feeling an arterial pulse, locating the heartbeat, or watching a muscle in action.

1.3 Anatomical terminology

Anatomy and medicine share an internationally agreed vocabulary, most of it drawn from Latin and Greek. The point of this shared, precise language is to remove ambiguity so that health professionals and scholars anywhere in the world can communicate without confusion. In practice a clinician must also know the everyday names for structures, so that patients can be understood and given explanations they follow. The terms can feel forbidding at first, but they start to make sense once their roots are learned.

1.4 The anatomical position

Every positional term is defined against a single agreed starting posture, the **anatomical position**. A person is in the anatomical position when standing upright with the feet together and the toes pointing forward, the arms by the sides with the palms turned forward, the head facing forward, the mouth closed and the expression neutral. This reference posture is used worldwide, so that a description means the same thing to everyone.

Lying position	Definition
Supine	Lying in the anatomical position but face-up.
Prone	Lying in the anatomical position but face-down.

1.5 The anatomical planes

Descriptions of location rely on four imaginary flat surfaces passing through the body. Each plane slices the body in a defined direction and is the basis for how sections and images are oriented.

Plane	How it passes through the body
Median	A vertical plane running lengthwise down the exact midline, dividing the body into equal left and right halves.
Sagittal	A vertical plane running parallel to the median plane (i.e. off-midline), giving unequal left and right portions.
Coronal (frontal)	A vertical plane at right angles to the median plane, dividing the body into front (anterior) and back (posterior).

Plane	How it passes through the body
Transverse	A horizontal plane at right angles to both the median and coronal planes, dividing the body into upper (superior) and lower (inferior) parts.

1.6 Terms of relationship

A set of paired opposites describes where one structure lies relative to another, always assuming the body is in the anatomical position. Learning them in pairs makes them easier to recall and to apply.

Term	Meaning	Note / example
Superficial vs deep	Nearer the surface vs further from the surface.	With 'intermediate' for a middle position; used to relate structures such as muscle to skin and bone.
Medial vs lateral	Nearer the median plane vs further from the median plane.	The little (5th) finger is medial; the thumb is lateral. Not the same as internal/external.
Anterior (ventral) vs posterior (dorsal)	Toward the front vs toward the back.	'Ventral' and 'dorsal' are equivalents used mainly by embryologists and neuroanatomists.
Superior vs inferior	Nearer the top of the skull (vertex) vs nearer the soles of the feet.	'Cranial' (toward the head) and 'caudal' (toward the tail) are the embryological equivalents.

A COMMON TRAP

Medial and lateral are not the same as internal and external. Medial and lateral describe distance from the midline plane; internal and external describe distance from the centre of an organ or cavity. Keeping the two pairs separate avoids a frequent exam error.

2. Movements of the Body

Movement happens at **joints** — the sites where two or more bones or cartilages meet and articulate. Every movement term is described from the anatomical position. Broadly, joint movements are grouped as angular and circular, and a further set of special movements is reserved for regions the general terms do not fit.

2.1 Angular movements

Angular movements increase or decrease the angle between the articulating bones. The four basic angular movements are flexion, extension, abduction and adduction.

Movement	Description
Flexion	Bending that decreases the joint angle. At the elbow the upper limb bends anteriorly; at the knee the lower limb bends posteriorly.
Extension	Straightening that increases the joint angle, returning the part toward the anatomical position — usually posteriorly, but anteriorly at the knee. A fully extended joint sits at about 180° (the ankle is an exception at roughly 90°).
Hyperextension	Extension carried beyond the anatomical position, giving an angle greater than 180°. It can cause injury — for example the 'whiplash' hyperextension of the neck in a rear-end collision.
Dorsiflexion	Flexion at the ankle, as when walking uphill or lifting the toes off the ground.
Plantarflexion	Turning the foot or toes toward the sole, as when standing on tiptoe. (When the foot is 'extended' it is in fact plantarflexed.)
Abduction	Moving a part away from the median plane in the coronal plane, as in raising a limb away from the side. For the digits it means spreading them apart from the middle finger (or 2nd toe).
Adduction	Moving a part toward the median plane, as in bringing a limb back to the side. For the digits it means drawing them toward the middle finger.

2.2 Circular movements

At joints that allow circular motion, a rounded or oval surface on one bone fits into a matching depression on another. The two basic circular movements are rotation and circumduction.

Movement	Description
Rotation	Turning a part around its own long axis, as in turning the head to the side. Medial rotation carries the anterior surface of a limb toward the midline; lateral rotation carries it away.
Circumduction	A combined, cone-shaped movement blending flexion, extension, abduction and adduction so that the far end of the part sweeps out a circle. It is possible only where all four component movements are — for example at the shoulder, hip, wrist, and the metacarpophalangeal and metatarsophalangeal joints.

2.3 Special movements

Some regions need their own terms because the general ones do not describe their motion well.

Movement	Description
Inversion vs eversion	Inversion turns the sole toward the midline (and is accompanied by plantarflexion when full); eversion turns the sole away from the midline (accompanied by dorsiflexion when full). Both are used clinically to describe developmental abnormalities.

Movement	Description
Opposition vs reposition	Opposition brings the pad of the thumb to the pad of another finger — the movement used to pinch, to button a shirt or to lift a cup by its handle. Reposition returns the thumb to the anatomical position.
Protrusion vs retrusion	Protrusion moves a part forward (as in jutting the jaw out); retrusion moves it back (tucking the chin in). The related terms protraction and retraction are used mostly for forward and backward movements of the shoulder.
Elevation vs depression	Elevation raises a part superiorly, as in shrugging the shoulders; depression lowers it inferiorly, as in letting the shoulders drop.
Pronation vs supination	Of the forearm and hand: pronation rotates the radius medially so the palm faces posteriorly (or downward when the elbow is flexed); supination rotates the radius laterally so the palm faces anteriorly — back into the anatomical position.

PART TWO · THE SKELETAL SYSTEM

3. The Skeletal System

The study of the structure and function of the skeleton and its bones is called **osteology**. Bones are the basic units of the skeleton: together they form the framework that shapes the body and bears its weight. A newborn has roughly 350 bones, but many fuse during childhood and adolescence, leaving the adult skeleton with about **206 bones**.

3.1 Functions of the skeleton

The skeleton performs both mechanical roles — support, protection and leverage for movement — and metabolic roles, namely blood-cell formation and the storage of fat and minerals.

Function	Explanation
Support	The bones form a rigid frame to which softer tissues and organs attach. Remarkably, the 206 bones support a mass of muscle and organs that can weigh about five times as much as the bones themselves.
Protection	The skull and vertebral column shield the brain and spinal cord; the rib cage guards the heart, lungs, great vessels, liver and spleen; and the pelvic girdle protects the pelvic organs. Even the blood-forming tissue is protected inside spongy bone.
Body movement	Bones act as levers and the joints as pivots: when muscles that anchor to bone contract, the bones move.
Fat storage	Lipid is stored as yellow bone marrow in the medullary cavity of certain bones.
Haemopoiesis	Blood-cell formation occurs in red bone marrow inside some bones. In infants the liver and spleen make red cells, but as bones mature the marrow takes over — producing on the order of 2.5 million red cells per second to replace those worn out and destroyed by the liver.

Function	Explanation
Mineral storage	The inorganic bone matrix is mainly calcium and phosphorus, which account for roughly two-thirds of bone's weight and give it firmness. About 95% of the body's calcium and 90% of its phosphorus are held in the bones and teeth.

3.2 Divisions of the skeleton

The skeleton is divided into two parts. The **axial skeleton** forms the central framework — the skull, which protects the brain; the vertebral column, which protects the spinal cord and supports the body; and the sternum and ribs (the rib cage), which protect the lungs and heart. The **appendicular skeleton** comprises the bones of the limbs together with the pectoral and pelvic girdles that anchor those limbs to the axial skeleton.

3.3 Types of bone

Bones are classified by their shape, and shape reflects the mechanical role each bone plays.

Type	Description and examples
Long	Tubular, with a hollow shaft and two ends, and containing a medullary cavity — the bones of the limbs.
Short	Roughly cube-shaped and found only in the wrist and foot — the carpals and tarsals.
Flat	Thin and flat — the bones of the skull, the sternum, the ribs and the pelvic bones.
Irregular	Bones of no simple shape, such as the facial bones and the vertebrae; many also hold blood-forming marrow.
Sesamoid	Bones that develop within a tendon; the largest example is the patella, or kneecap.

3.4 The axial skeleton

The vertebral column

In adults the vertebral column is about 72–75 cm long. It begins as 33 vertebrae arranged in five regions. The sacral and coccygeal vertebrae fuse into single bones, so although 33 vertebrae are present at first, movement occurs across only 24 of them.

Region	Number	Note
Cervical	7	Neck region.
Thoracic	12	Articulate with the ribs.
Lumbar	5	Lower back; bears much of the body's weight.
Sacral	5 to 1	Fuse to form the sacrum.
Coccygeal	4 to 1	Fuse to form the coccyx.

The skull

The skull is built from two sets of bones: the **cranial bones**, which form the braincase (cranium), and the **facial bones**, which support the eyes and nose and frame the mouth. The facial skeleton is made up of 14 irregular bones. On an anterior view the frontal, parietal, temporal, nasal, lacrimal, maxilla, mandible and vomer can be identified; a lateral view adds the occipital, sphenoid and zygomatic bones and the temporomandibular joint (TMJ).

Auditory ossicles and the hyoid

Each middle ear contains three tiny **auditory ossicles** (“ear bones”) that transmit sound. The **hyoid bone** sits above the larynx (voice box) and below the mandible (jawbone); it supports the tongue and assists in swallowing, and is unusual in articulating with no other bone.

The rib cage

The rib cage forms the bony and cartilaginous framework of the thorax. It joins the thoracic vertebrae behind and comprises 12 pairs of ribs, the flat sternum in front, and the costal cartilages that connect the ribs to the sternum.

3.5 The appendicular skeleton

The appendicular skeleton consists of the limb bones and the girdles that attach them to the axial skeleton.

Component	Bones and role
Pectoral girdle	The paired scapulae (shoulder blades) and clavicles (collarbones) are its appendicular parts, with the sternum as the axial part. Its main role is to anchor the muscles that move the arm and forearm.
Upper limb	Humerus in the arm; radius and ulna in the forearm; then the carpals, metacarpals and phalanges of the hand.
Pelvic girdle	The two hip bones are its appendicular parts and the sacrum its axial part. The hip bones join anteriorly at the pubic symphysis and posteriorly at the sacrum; the girdle transmits body weight through the vertebral column and protects the pelvic organs.
Lower limb	Femur (thighbone) in the thigh; tibia (shinbone) and fibula in the leg; then the tarsals, metatarsals and phalanges of the foot. The patella (kneecap) lies on the front of the knee.

4. Joints (Articulations)

One of the skeleton's key jobs is to allow movement — yet it is not the bones themselves that move but the unions between them, called **joints** or articulations. Joints vary widely: some permit no movement, some only slight movement, and others move freely. The science of joints is **arthrology**, concerned with their structure, classification, function and any dysfunction.

4.1 Classification of joints

Joints are classified by the material that unites the articulating bones, giving three main types — fibrous, cartilaginous and synovial.

Fibrous joints

In fibrous joints the bones are held together by fibrous connective tissue, giving unions that range from rigid to slightly movable. There are three kinds:

Fibrous joint	Description
Sutures	Found only in the skull, where a thin layer of dense irregular connective tissue binds the bones. They form at about 18 months of age, replacing the pliable fontanels of the infant skull. Types include serrate, squamous and plane sutures.
Syndesmoses	Bones joined by collagenous fibres or sheets called interosseous ligaments — as between the radius and ulna in the forearm and the tibia and fibula in the leg, and between adjacent vertebral processes. They allow slight movement, e.g. during rotation of the forearm or leg.
Gomphoses	The peg-in-socket joint between a tooth and its bony socket (the dental alveolus), where the tooth root attaches to the periodontal ligament.

Cartilaginous joints

Cartilaginous joints permit limited movement in response to twisting or compression. There are two kinds:

Cartilaginous joint	Description
Symphyses	The bone surfaces are capped with hyaline cartilage that becomes infiltrated with collagen to form a pad of fibrocartilage, which cushions the joint and allows limited movement — as at the pubic symphysis and the intervertebral discs.
Synchondroses	Joints in which hyaline cartilage lies between the articulating bones — for example where the ribs join the sternum.

Synovial joints

Synovial joints are the freely movable joints. Each is enclosed by a joint capsule containing lubricating synovial fluid. They are subclassified by the shape of the articulating surfaces and the movements those shapes allow.

Synovial type	Movement and example
Plane (gliding)	Nearly flat surfaces that allow side-to-side and back-and-forth sliding with slight rotation — the simplest movement. Examples: intercarpal, intertarsal and sternoclavicular joints.
Hinge	A uniaxial (monoaxial) joint like a door hinge, allowing only flexion and extension in one plane — the elbow and knee.
Pivot	Movement limited to rotation about a central axis, where a rounded surface fits into a ring or depression — the atlas–axis joint that turns the head.
Condyloid	An oval surface fitting a matching cavity, as in a knuckle (metacarpophalangeal) joint.
Saddle	Reciprocally saddle-shaped surfaces, as where the trapezium meets the first metacarpal — the carpometacarpal joint responsible for the opposable thumb, a hallmark of primate anatomy.

Synovial type	Movement and example
Ball-and-socket	A rounded head fitting a cup-like cavity, giving the greatest, multiaxial range of movement — the shoulder (glenohumeral) and hip (coxal) joints.

PART THREE · THE MUSCULAR SYSTEM

5. The Muscular System

Myology is the branch of anatomy that describes the structure and function of muscle. Muscles belong to the group of structures called **effectors** — tissues that respond to a stimulus, either directly or indirectly. There are more than 600 skeletal muscles, each technically an organ, and together they make up about **40% of body weight**.

5.1 Functions of muscle

Muscles perform three principal functions.

Function	Explanation
Movement	The most obvious role — moving the body or its parts in walking, running, writing, chewing and swallowing. Even the eyeball and the ear ossicles have muscles, and skeletal muscle also drives breathing and helps move internal fluids.
Heat production	Body temperature is kept remarkably constant. Cellular metabolism releases heat, and because muscle is such a large, continuously active fraction of the body it is the main source of body heat — rising sharply during hard exercise.
Posture and support	Muscles maintain posture, stabilise the flexible joints and support the internal organs. Some postural muscles work even when you feel relaxed — while sitting, muscles at the back of the neck balance the weight of the head at the atlanto-occipital joint.

5.2 Properties of muscle tissue

All muscle tissue shares four basic properties that make contraction possible and repeatable.

Property	Meaning
Irritability (excitability)	Sensitivity to stimuli from nerve impulses.
Contractility	The ability to respond to a stimulus by shortening (contracting lengthwise).
Extensibility	Once relaxed, fibres can be stretched — even beyond their resting length — by the pull of an opposing muscle, readying them for the next contraction.
Elasticity	After being stretched, fibres tend to recoil to their original resting length.

5.3 Types of muscle

Three types of muscle are recognised by their location, structure and control.

Type	Also called	Key features
Skeletal	Striated / voluntary	Innervated by the voluntary nervous system, so it is under conscious control; shows stripes (striations) under the microscope. Most attach to bone directly or through tendons — others to cartilage, ligaments, fascia, skin (the facial muscles), organs (the eye muscles) or mucous membranes (the intrinsic tongue muscles).
Smooth	Visceral / involuntary	Lines the cavities and hollow organs; innervated by the autonomic nervous system, so it is involuntary; has no striations. Found in the middle coat of blood-vessel walls, the wall of the digestive tract, and the skin. It controls most involuntary actions.
Cardiac	—	Forms the myocardium, the muscular wall of the heart (with some in the walls of the aorta, pulmonary vein and superior vena cava). Striated like skeletal muscle, but involuntary — its contraction is set intrinsically by a pacemaker of specialised cardiac fibres.

ORIGIN VS INSERTION

A muscle's attachments are described as its **origin** and its **insertion**. The origin is usually the proximal attachment that stays relatively fixed during contraction, while the insertion is usually the distal attachment that moves. Knowing which end moves tells you what the muscle actually does.

5.4 Internal structure of muscle

A skeletal muscle is built from hundreds of muscle fibres, each between about 1 and 40 mm long. Inside every fibre run finer threads called **myofibrils**. Under the microscope a myofibril shows alternating light and dark bands, and because the bands of neighbouring myofibrils line up side by side, the whole fibre appears crossed by continuous stripes — the striations that give skeletal and cardiac muscle their name.

Banding of the myofibril

The banding pattern has a regular internal architecture. Each dark band carries a paler central zone, the **H-zone**, flanked by darker regions, and within the H-zone lies a dark line, the **M-membrane**. Each light band is crossed in the middle by another dark line, the **Z-membrane**. The stretch from one Z-membrane to the next is the **sarcomere** — the basic contractile unit of the myofibril.

Landmark	Location / meaning
Dark band	Contains the thick filaments.

Landmark	Location / meaning
Light band	Contains the thin filaments.
H-zone	Paler region in the middle of the dark band.
M-membrane	Dark line at the centre of the H-zone; anchors the thick filaments.
Z-membrane	Dark line crossing the middle of the light band; anchors the thin filaments.
Sarcomere	The unit from one Z-membrane to the next — the basic building block of the myofibril.

5.5 How muscle contracts

Each myofibril contains two kinds of filament: **thick** filaments, confined to the dark band, and **thin** filaments, confined to the light band. These are built from two proteins — **myosin** in the thick filaments and **actin** in the thin filaments. When actin and myosin are placed together in a solution of **ATP**, contraction occurs; actin, myosin and ATP are therefore the three basic requirements for contraction.

Contraction happens when the thick and thin filaments slide between one another. The thin filaments are held together by the Z-membrane and the thick filaments by the M-membrane, so the filaments slide as an organised unit and the orderly banding is preserved throughout the process — the essence of the sliding-filament mechanism.

PART FOUR · THE CIRCULATORY SYSTEM

6. The Cardiovascular System

The cardiovascular system is the body's transport network, made up of the **heart** and the **blood vessels**. Through it the body moves the materials cells need and carries away the wastes they produce.

6.1 Functions

Function	Explanation
Transport	Carries oxygen, nutrients and hormones to the cells and removes wastes such as carbon dioxide and urea.
Protection	White blood cells carried in the blood defend the body against infection and disease.
Regulation	Helps regulate body temperature, pH and fluid balance.

6.2 The heart

The heart is a conical, hollow, muscular organ that works without pause throughout life. It is a little larger than a clenched fist and weighs about 300 g. It sits in the chest just behind the sternum and between the two lungs, wrapped in a tough two-layered membrane, the **pericardium**. Its wall is built largely of **cardiac**

muscle, which contracts and relaxes rhythmically and can work continuously without tiring.

At rest the heart beats about **72 times a minute**, pumping roughly 14,200 litres of blood a day; during strenuous effort the rate rises to about 100 beats a minute, delivering more oxygen and nutrients to the tissues.

The wall of the heart

The heart wall has three layers. From outside in these are the **pericardium**, the **myocardium** and the **endocardium**. The pericardium itself consists of two sacs — an outer fibrous sac and an inner double layer of serous membrane.

6.3 Chambers, valves and great vessels

The heart has four chambers: two thin-walled upper chambers, the **atria** (also called auricles), and two thick-walled lower chambers, the **ventricles**. A muscular wall, the **septum**, completely separates the right side from the left. Blood returning from the body enters the right atrium, while blood returning from the lungs enters the left atrium; rings of muscle within the veins control these openings.

Structure	Role
Superior & inferior venae cavae	Two large veins that return dark, deoxygenated blood from the body (except the lungs) into the right atrium.
Pulmonary veins	Return bright, oxygenated blood from the lungs into the left atrium.
Tricuspid valve	Guards the right atrium–ventricle opening; has three flaps (cusps) tethered by chordae tendineae to the ventricle wall.
Bicuspid (mitral) valve	Guards the left atrium–ventricle opening; like the tricuspid but with two flaps.
Pulmonary artery	Leaves the right ventricle and divides to the two lungs, carrying deoxygenated blood.
Aorta	Leaves the left ventricle and branches to supply oxygenated blood to the whole body except the lungs.

WHY THE LEFT VENTRICLE IS THICKER

The wall of the left ventricle is at least three times thicker than that of the right. The left ventricle must push blood through the long systemic circulation to the whole body, whereas the right ventricle only drives blood through the much shorter pulmonary circulation to the nearby lungs — so the left needs far more muscular force.

6.4 The heart sounds

Through a stethoscope a normal heart makes a repeating “lub-dub” sound. The first sound, **“lub,”** is produced as the atrioventricular valves (bicuspid and tricuspid) close at the start of ventricular contraction (systole). The second sound, **“dub,”** is produced as the semilunar valves (aortic and pulmonary) close at the end of systole.

6.5 Blood vessels

There are three kinds of blood vessel: **arteries**, **veins** and **capillaries**.

Arteries

Arteries carry oxygenated blood away from the heart to the tissues — the one exception being the pulmonary artery, which carries deoxygenated blood to the lungs. Delivery to the tissues is completed by the smallest arteries, the **arterioles**. The wall of an artery has three coats, or tunicae, from inside out: the **tunica intima**, the **tunica media** and the **tunica adventitia**. Arteries are classified by the thickness and make-up of these coats, especially the tunica media.

Artery type	Features
Elastic (conducting)	The largest arteries, with elastic fibres in their walls that let them expand as the heart contracts and recoil between beats — e.g. the aorta and its arch branches (brachiocephalic trunk, left common carotid).
Muscular (distributing)	Medium-sized arteries with smooth muscle in the tunica media, supplied by sympathetic nerves so they can constrict or dilate; they distribute blood throughout the body.
Arterioles	The smallest distributing arteries, almost entirely muscular in their coats, delivering blood into the capillaries.

ANASTOMOSES OF ARTERIES

An **anastomosis** is a connection between arteries, and two kinds matter clinically. In an **actual anastomosis** the arteries meet end to end, so if cut they bleed from both ends — for example the uterine and intercostal arteries. In a **potential anastomosis** the link is through terminal arterioles, which given time can enlarge to divert an adequate supply — as around joints.

Veins

Veins carry blood from the tissues (from the capillaries) back to the heart. They carry deoxygenated blood, except for the pulmonary vein, which carries oxygenated blood. A vein wall has the same three coats as an artery — tunica intima, media and adventitia — and veins come in three sizes: small (venules), medium and large.

Capillaries

Capillaries are the smallest vessels: simple endothelial tubes that link the arterial and venous sides of the circulation. They are arranged in networks called **capillary beds** lying between the arterioles and the venules. Blood is delivered to a capillary bed by arterioles and drained from it by venules, and it is across these thin walls that the actual exchange of gases, nutrients and wastes takes place.

PART FIVE · THE INTEGUMENTARY SYSTEM

7. The Skin (Integumentary System)

The **integument** consists of the skin — the epidermis and the dermis — together with its appendages: the sweat glands, sebaceous glands, hair and nails. It is regarded as the **largest organ** of the body, covering more than 7,600 cm^2 (about 3,000 in^2) in an average adult, and the heaviest, making up roughly **7% of total body weight**.

7.1 Thickness and clinical appearance

Skin varies in thickness, averaging about 1.5 mm. It is thickest — up to about 6 mm — on areas exposed to wear such as the soles and palms, and thinnest — around 0.5 mm — on the eyelids, external genitalia and eardrum. Its texture likewise ranges from the rough, calloused skin of the elbows and knuckles to the soft skin of the eyelids and nipples.

Skin appearance is clinically valuable because it hints at underlying problems: pale skin may signal shock, while red, flushed, over-warm skin may point to fever or infection. A rash may reflect allergy or local infection, abnormal texture may follow glandular or nutritional disturbance, and even chewed fingernails can be a clue to emotional stress.

7.2 Functions of the skin

Function	Explanation
Protection	A barrier against microorganisms, water and excess ultraviolet light. Oily secretions form an acidic surface film (pH 4.0–6.8) that waterproofs the body and slows the growth of most pathogens, and the protein keratin in the epidermis adds further waterproofing.
Vitamin D synthesis	The skin's most important metabolic role: ultraviolet light drives the production of vitamin D3 (1,25-dihydroxycholecalciferol), essential for calcium metabolism and bone formation.
Sensation	Receptors in the skin detect heat, cold, touch, pressure, vibration and pain.
Thermoregulation	Achieved through thermoreceptors and the sweat glands.
Hydroregulation	Helps control water loss from the body.
Social & psychological roles	Serves as a sexual and psychological attractant, contributes to appearance and social acceptance, and enables facial expression and non-verbal communication.

7.3 Layers of the skin

The skin has three main layers: the superficial **epidermis**, the deeper **dermis**, and beneath them the **subcutis (hypodermis)**.

Epidermis

The epidermis is the superficial, protective layer. It is a stratified squamous epithelium of ectodermal origin, ranging from about 0.007 to 0.12 mm thick, and it contains no blood vessels. From deep to superficial it is built of four or five layers.

Layer (deep to superficial)	Feature
Stratum basale	The deepest layer, where new cells and melanin are produced.
Stratum spinosum	A layer in which Langerhans (immune) cells are most easily seen.
Stratum granulosum	A granular layer in the course of keratinisation.
Stratum lucidum	A clear layer present only in thick skin — the lips, palms and soles.
Stratum corneum	The superficial layer of several sheets of keratinised (dead) cells.

Cells of the epidermis

Cell	Role
Keratinocytes	The most numerous cells; produce the keratin proteins that toughen and waterproof the skin.
Melanocytes	Specialised epithelial cells that make the pigment melanin (packaged as melanosomes), which shields against ultraviolet radiation and gives skin its colour.
Langerhans cells	Bone-marrow-derived immune cells carried to the skin by the blood; act as antigen-presenting cells and are most easily recognised in the stratum spinosum.
Merkel (tactile) cells	Found in the basal epidermis; work with closely associated nerve fibres as sensory touch receptors.
Granular dendrocytes	Non-pigmented, protective macrophage-like cells scattered through the stratum basale that ingest bacteria and debris.

Dermis

The dermis lies beneath the epidermis. Its collagen and elastic fibres are laid down in definite patterns, creating lines of tension and giving the skin its tone and texture. In youth the collagen and elastin keep the skin tight and firm, but with age — and especially with sun exposure — these fibres in the upper dermis degenerate, so the skin loses texture and wrinkles. The dermis also houses the skin appendages, the oil glands, most of the skin's blood supply, and its nerves and sensory endings. It has two layers:

Layer	Description
Papillary (stratum papillarosum)	In contact with the epidermis and forming about one-fifth of the dermis; its projections (papillae) form the base of the friction ridges of the fingers and toes.
Reticular (stratum reticularosum)	The deeper, thicker layer, with dense, regularly arranged fibres forming a tough, flexible and distensible meshwork. Overstretching tears it, and the repair leaves a white streak — a stretch mark, or linea albicans.

Subcutis (hypodermis)

Also called the subcutaneous tissue, the hypodermis holds more adipose (fat) tissue than the dermis and fastens the skin to the underlying muscles and other structures. It is on average about 8% thicker in females than in males. It serves as an energy reservoir, storing lipids, and as thermal insulation.

7.4 Skin colour

Normal skin colour comes from a combination of three pigments — **melanin**, **carotene** and **haemoglobin**. Melanin is a brown-black pigment made by the melanocytes of the stratum basale; all people of similar size have about the same number of melanocytes, so racial differences in colour come from how much melanin is made and how it is distributed. Carotene is a yellowish pigment found in plant foods such as carrots; it was once thought to explain the yellow-tan skin of people of Asian descent, but that colouring is now known to be due to melanin. Haemoglobin is not truly a skin pigment — it is the oxygen-carrying pigment of red blood cells, and oxygenated blood flowing through the dermis lends the skin its pinkish tones.

7.5 Skin appendages

Hair

Humans are relatively hairless, with dense hair only on the scalp, face, pubis and axillae; the palms, soles, lips, nipples and parts of the genitalia are hairless. Hair is chiefly protective — scalp and eyebrow hair guards against sunlight, eyelashes and nasal hairs against airborne particles, and scalp hair against minor injury. Each hair has a **shaft** (the visible but dead part above the surface), a **root**, and a **bulb** (the enlarged base within the follicle). Scalp hair grows about 1 mm every three days before entering a resting phase; a hair's lifespan ranges from 3–4 months for an eyelash to 3–4 years for a scalp hair, each lost hair being replaced by a new one that pushes the old out. Between 10 and 100 hairs are shed daily. Small bundles of smooth muscle, the **arrector pili**, attach to the follicle and the papillary dermis; when they contract they raise the hairs and dimple the skin into “goose bumps.”

Glands

Gland	Description
Sebaceous (oil) glands	Associated with hair follicles, these holocrine glands secrete sebum onto the hair shaft. If their ducts become blocked they may become infected, producing acne; sex hormones regulate sebum output, so overactivity in the teenage years can cause serious acne.
Sweat glands	Coiled, tubular glands that excrete perspiration — water, salts, urea and uric acid — onto the skin, both for evaporative cooling and to excrete certain wastes. They are most numerous on the palms, soles, axillary and pubic regions and the forehead, and are of two types, eccrine and apocrine.

Nails

The nails on the fingers and toes are formed from the compressed outer layer (stratum corneum) of the epidermis. Their hardness comes from dense keratin fibrils running parallel between the cells. Nails protect the tips of the digits, and the fingernails also help in grasping and picking up small objects.

7.6 Applied anatomy of the skin

Condition	Description
Acne	An inflammatory condition of the sebaceous glands, influenced by gonadal hormones and therefore common in puberty and adolescence; seen as pimples and blackheads on the face, chest and back.
Melanoma	A cancerous tumour arising from proliferating melanocytes within the epidermis.
Hirsutism	Excessive body and facial hair, especially in women. It may be genetic (as in some ethnic groups) or due to a metabolic — usually endocrine — disorder, and can appear with the hormonal changes of menopause. Treatments include hormonal therapy and electrolysis to destroy selected follicles.
Albinism	A group of genetic disorders giving little or no pigment in skin, hair or eyes, caused by melanocytes being unable to make melanin — through absent tyrosinase activity or an inability to take up tyrosine.
Vitiligo	A disorder in which melanocytes are destroyed, thought to be secondary to autoimmune dysfunction, causing depigmented areas — macules if under 1 cm, patches if larger.
Ichthyosis	From the Greek ichthys, 'fish': excessive keratinisation causing dryness and fish-scale-like scaling that may cover the whole body.
Alopecia	Absence or loss of scalp hair. Congenital alopecia follows failure of follicles to develop or poor-quality hair; pressure alopecia can result from sustained pressure on the scalp, as in infants who lie on their backs.

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This guide was compiled and rewritten in original prose from general anatomy course materials, reorganised into a logical, self-contained revision resource. Facts, values and mechanisms have been preserved; wording is our own. If you spot an error, or want a topic expanded, get in touch through everythingabuad.com so we can keep improving it.